Cyber Defense Enablement

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Introduction (Why → How → What → So What)

Why Cyber Defense Enablement Exists

Technology solutions deployed within HCSC operate in an environment where cyber-attacks are continuous, adaptive, and often highly sophisticated. Attackers do not distinguish between infrastructure, applications, or vendors; they exploit the weakest point available to gain access to enterprise systems, sensitive data, and critical operations. HCSC is under constant cyber-attack 24 hours per day, 7 days per week, 365 days of the year.

While the Secure-by-Design model ensures that technology solutions are built with appropriate security controls, security controls alone do not stop attacks. Every system and application will eventually be probed, tested, and targeted by malicious actors.

When the attack against your project's solution occurs, the organization must be able to:

- Detect malicious activity
- Investigate the scope of the attack
- Contain the threat
- Restore normal operations
- Preserve evidence and understand the root cause

These capabilities are required by law & contractual agreements, and they are the responsibility of the Cyber Defense & Investigation (CD&I) team within the Information Security organization. The CD&I teams' mission is to defend HCSC and its members from all forms of cyber threat, internal and external, foreign and domestic.

For CD&I to perform this mission, technology solutions must be designed and deployed in a way that enables cyber defense operations. Without this enablement, security teams cannot effectively monitor, detect, investigate, or respond to threats affecting the system.

Cyber Defense Enablement ensures that every technology solution deployed within the enterprise can be actively defended against cyber threats throughout its lifecycle.

How Cyber Defense Enablement Works

Cyber Defense Enablement integrates technology solutions with the Cyber Defense ecosystem, which includes monitoring platforms, detection analytics, investigative tools, and incident response capabilities operated by CD&I.

This integration ensures that:

- Security telemetry from the solution is collected and monitored
- Threat detection capabilities are implemented

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- Investigative visibility is available when suspicious activity occurs
- Incident response actions can be executed quickly and effectively

Cyber Defense Enablement is implemented through a standard work package delivered as part of the project lifecycle. This work package defines the architecture, engineering integration, detection capabilities, and operational procedures required to defend the technology solution.

Because CD&I operates as an operational defense organization and does not maintain project delivery resources, the project sponsoring the technology solution must plan, fund, and execute the required integration work with support from Information Security Architecture, Information Security Engineering, Security Operations, and external specialists as required.

What Cyber Defense Enablement Delivers

The Cyber Defense Enablement work package ensures that technology solutions are deployed with the operational capabilities required for enterprise defense.

These capabilities include:

- Security telemetry generation
- Log collection and integration with enterprise monitoring platforms
- Threat detection analytics and automation
- Investigation and forensic visibility
- Incident response procedures and playbooks
- Operational monitoring runbooks
- Metrics and dashboards supporting defense operations

When completed, Cyber Defense Enablement ensures that the technology solution can be continuously monitored and defended by CD&I once it enters production.

So What: Why This Matters

Failure to enable Cyber Defense creates significant operational and enterprise risk. Systems that cannot be monitored and defended create blind spots that attackers can exploit to gain and maintain access to the enterprise environment.

What Will Happen Without Cyber Defense Enablement

If Cyber Defense Enablement is not implemented, the following outcomes are certain to occur:

- Security incidents affecting the system will not be detected in a timely manner
- Attackers may operate within the system for extended periods without detection

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- Security teams will lack the telemetry required to investigate incidents
- Incident response efforts will be delayed or ineffective
- The organization will have limited ability to determine the scope or impact of a breach
- When the incident occurs, security teams may be forced to assume the worst-case scenario, which may require:
  - full system shutdown
  - emergency containment actions
  - extensive forensic investigation
  - regulatory and legal escalation

These actions can significantly disrupt business operations and will result in material financial and reputational impact.

What Might Happen Without Cyber Defense Enablement

If Cyber Defense Enablement is not implemented, the following outcomes may occur:

- Attackers may be able to:
  - establish persistent access within the environment
  - move laterally to other enterprise systems
  - escalate privileges and compromise administrative accounts
  - access or exfiltrate sensitive data
  - manipulate or disrupt critical business services
  - deploy ransomware or destructive malware

In modern cyber incidents, attackers frequently remain undetected within enterprise environments for extended periods while they expand their access and prepare for larger attacks.

Systems that cannot be monitored by Cyber Defense effectively provide an unguarded entry point into the enterprise environment.

The Enterprise Standard

For these reasons, all technology solutions deployed within the HCSC environment must be designed and implemented in a way that enables Cyber Defense operations.

This requirement ensures that:

- the enterprise maintains visibility into system activity
- threats can be detected and investigated
- incidents can be contained and resolved
- enterprise risk can be effectively managed

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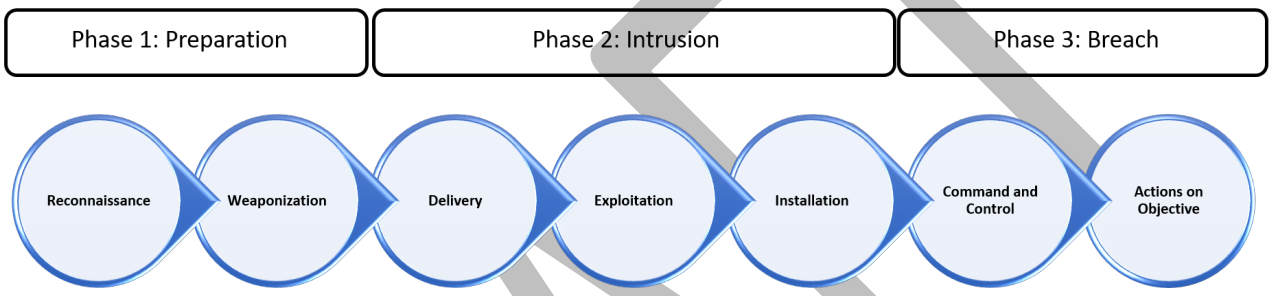
A system that cannot be monitored, detected, investigated, and defended by Cyber Defense is not operationally defensible and will not be deployed to production.

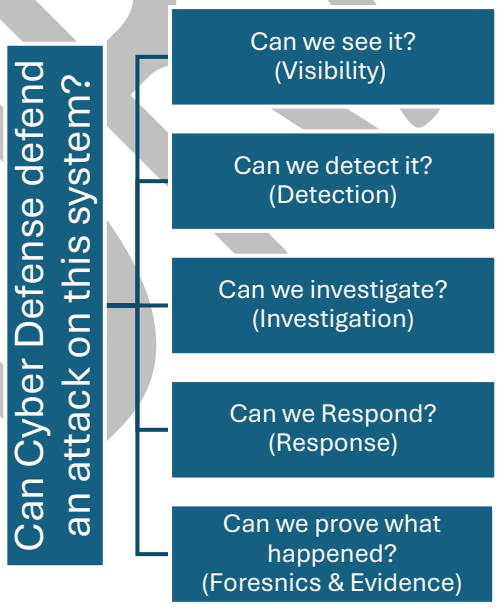
Cyber Defense Enablement ensures that every system introduced into the enterprise environment strengthens, rather than weakens, the organization's ability to defend itself against cyber threats.

Why Log Collection Alone Does Not Enable Cyber Defense

All cyber attacks can be described as a sequence of actions the attacker takes to achieve their objective, this is known as a kill chain.



To defend against an attack, the chain must be broken. This requires awareness of the attack and ability to prevent a link in the chain from occurring.



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A common misconception is that integrating a system's logs into a Security Information and Event Management (SIEM) platform is sufficient to enable cyber defense. Logs answer only part of the final question: Can we prove what happened?

While log collection is an important component of security visibility, log aggregation alone does not provide the capabilities required to detect, investigate, and respond to real-world cyber attacks.

Effective cyber defense depends on the ability to answer a small set of critical operational questions about a technology solution:

- How would we detect an attacker using this system?
- How would we detect misuse of legitimate credentials?
- How would we investigate an incident affecting the system?
- How would we contain an active attack?
- How would we know if sensitive data was accessed or exfiltrated?

These questions reflect how real-world cyber attacks unfold and what cyber defense teams must be able to do to stop them. If a system cannot provide the telemetry, context, and operational controls required to answer these questions, it cannot be effectively defended. Simply forwarding logs to a central platform does not solve this problem. Logs going to the SIEM do not answer any of these questions.

Modern cyber attacks do not occur as single events that can be identified from an isolated log entry. Instead, attackers execute multi-stage campaigns that unfold across multiple systems, identities, and network paths over time.

Detecting and responding to attacks requires meaningful security telemetry, detection analytics, investigative visibility, and response capabilities designed into the system architecture so that Cyber Defense teams can identify and stop malicious activity before it results in enterprise impact.

A typical enterprise attack progression includes several stages:

1. Initial access (phishing, credential theft, exploitation of an exposed service)
2. Establishing persistence
3. Privilege escalation
4. Command and control communication
5. Lateral movement across systems
6. Data discovery and exfiltration
7. Impact actions such as ransomware or service disruption

Detecting and stopping attacks requires visibility across this entire sequence of activity, not simply the collection of application or system logs.

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If cyber defense visibility is limited to basic log collection, the organization may observe isolated technical events but will lack the context necessary to recognize coordinated malicious activity.

Effective cyber defense requires that technology solutions provide the right types of telemetry, in the right format, with sufficient context, and with the ability to correlate activity across systems and identities.

What Cyber Defense Actually Requires

Effective Cyber Defense requires real time contextual awareness. Security incidents are not detected by personnel manually reviewing logs or watching screens of raw log data. Modern cyber defense operations rely on the continuous analysis of security event telemetry in real time, where analytics, correlation rules, and behavioral detections evaluate activity across systems, identities, and networks to identify patterns consistent with malicious behavior. When these analytics identify potential threats, alerts are generated and delivered to Cyber Defense teams for triage, investigation, and response. Human analysts are therefore reacting to alerts produced by detection systems, not manually "monitoring logs." Raw logs serve a different purpose in the defense lifecycle: they provide evidentiary detail used during incident investigation to reconstruct attacker activity and understand scope and impact. While logs are an important data source for detection and investigation, the concept of "log monitoring" is largely a misconception. Cyber Defense teams monitor alerts derived from telemetry analytics, not streams of raw log data.

Logs are evidence after an attack. Cyber defense requires the ability to detect and stop the attack while it is happening.

To enable effective cyber defense operations, technology solutions must support several operational capabilities beyond simple log forwarding.

These include:

Security Telemetry

The system must generate meaningful security telemetry that captures authentication activity, administrative actions, API usage, network connections, and other events relevant to detecting malicious behavior.

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Identity Attribution

Cyber defense teams must be able to attribute actions to specific users, service accounts, or workloads. Without reliable identity context, it becomes impossible to distinguish legitimate activity from attacker activity.

Investigation Visibility

Telemetry must contain sufficient detail to allow security teams to reconstruct attacker activity across systems. This includes timestamps, source devices, network connections, and contextual information required to understand the sequence of events.

Operational Response Capability

Cyber defense teams must be able to take action to contain or mitigate an attack. This may include disabling accounts, blocking network communication, isolating systems, or revoking credentials.

Detection Engineering Integration

Cyber defense monitoring relies on detection analytics and correlation rules that identify suspicious activity patterns across multiple data sources. Technology solutions must provide telemetry that supports these detection capabilities.

The Reality of Modern Cyber Attacks

Attackers exploit the gaps that exist between systems and security controls. When cyber defense telemetry is incomplete or poorly integrated, attackers can operate within the environment without triggering alerts.

In many enterprise breaches, security logs existed but were insufficient to detect the attack or reconstruct the attacker's activity. The absence of meaningful telemetry often becomes apparent only after an incident occurs.

Organizations that rely solely on log collection frequently discover during a breach investigation that:

- the relevant activity was never logged
- the logs lacked the context needed to identify malicious behavior
- the logs were not retained long enough for investigation
- the logs were never correlated with other security signals

As a result, security teams may be forced to assume the worst-case scenario and conduct extensive containment and recovery efforts.

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The Principle of Defensible Systems

Cyber Defense Enablement ensures that technology solutions are deployed in a way that allows the enterprise to actively defend them against cyber threats.

This requires more than compliance with logging standards or integration with monitoring platforms. It requires that systems be designed with the operational needs of cyber defense in mind.

A technology solution is considered defensible only when Cyber Defense can:

- observe meaningful activity within the system
- detect malicious behavior affecting the system
- investigate suspicious activity with sufficient context
- respond quickly to contain an attack

Systems that do not provide these capabilities create operational blind spots that attackers can exploit.

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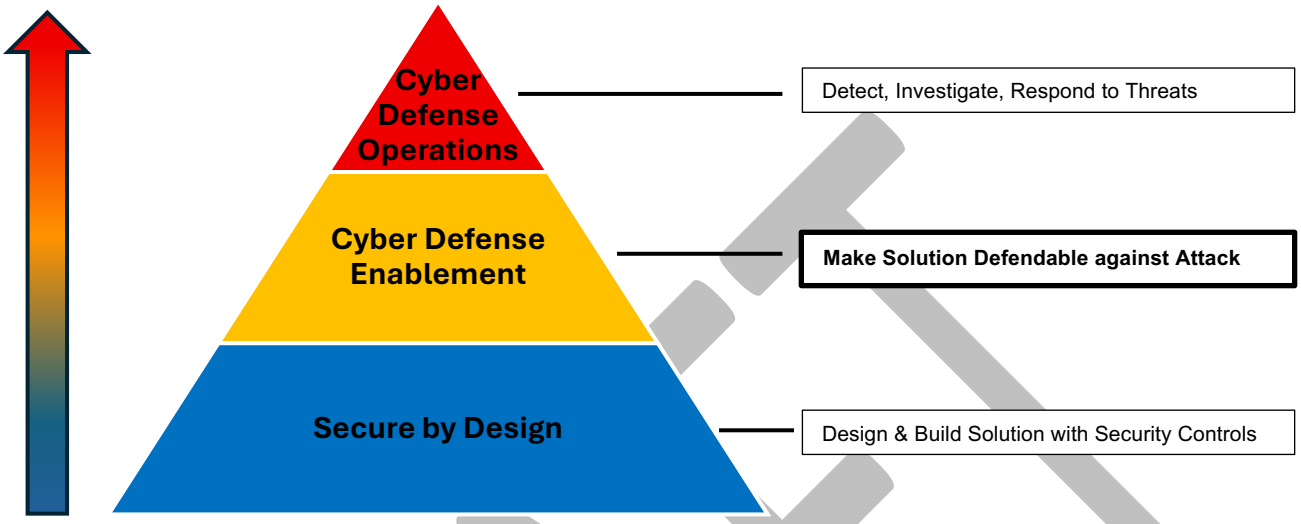
Cyber Defense Enablement Work Package Framework



Purpose

The Cyber Defense Enablement Work Package (CDWP) ensures that technology solutions deployed within the enterprise environment can be actively monitored, detected, investigated, and defended by the Cyber Defense & Investigation (CD&I) organization.

This work package establishes the architecture, engineering integrations, detection capabilities, and operational procedures required for CD&I to defend the technology solution against cyber threats.

Cyber Defense Enablement complements the Secure-by-Design model by ensuring that systems are not only built with required security controls but are also operationally defensible once deployed.

Cyber Defense Enablement does not create new security requirements. It ensures that existing security controls and enterprise cyber defense capabilities can effectively defend the technology solution.

Scope

The Cyber Defense Enablement Work Package applies to:

- new enterprise applications
- new infrastructure platforms
- new cloud services
- new SaaS platforms

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- new external integrations
- major architecture changes to existing systems

The work package ensures that each solution:

- generates meaningful security telemetry
- integrates with enterprise monitoring platforms
- supports threat detection analytics
- provides investigative visibility
- enables incident response containment actions

Engagement Model

Cyber Defense Enablement engagement may be required when a technology solution introduces new systems, platforms, services, or architectural changes that require monitoring, detection, investigation, or response capabilities from the Cyber Defense & Investigation (CD&I) organization. All projects delivering a technology solution must initiate Cyber Defense engagement early in the project lifecycle to determine whether Cyber Defense Enablement is required.

Engagement is initiated through submission of the Cyber Defense Enablement Engagement Form as part of project intake or architecture review.

Submission of the engagement form initiates Cyber Defense Enablement review and determines whether the Cyber Defense Enablement Work Package is required for the solution.

Engagement Form

To engage Cyber Defense Enablement, the project manager must complete the Cyber Defense Enablement Engagement Form provided in Appendix A and submit it to Cyber Defense & Investigation (CD&I). The form captures essential information about the solution architecture, exposure, integrations, and operational characteristics so that CD&I can evaluate how the solution will be monitored, detected, investigated, and defended once deployed. Completion of this form enables CD&I to determine the scope of Cyber Defense Enablement activities, identify required inputs from architecture and engineering teams, and estimate the effort and cost associated with enabling cyber defense for the solution. Projects may not proceed to production readiness review until the engagement form has been submitted and CD&I has completed its initial assessment.

Inputs, Outputs, and Internal Deliverables

Relationship Between Inputs and Outputs

The Cyber Defense Enablement process follows a simple model:

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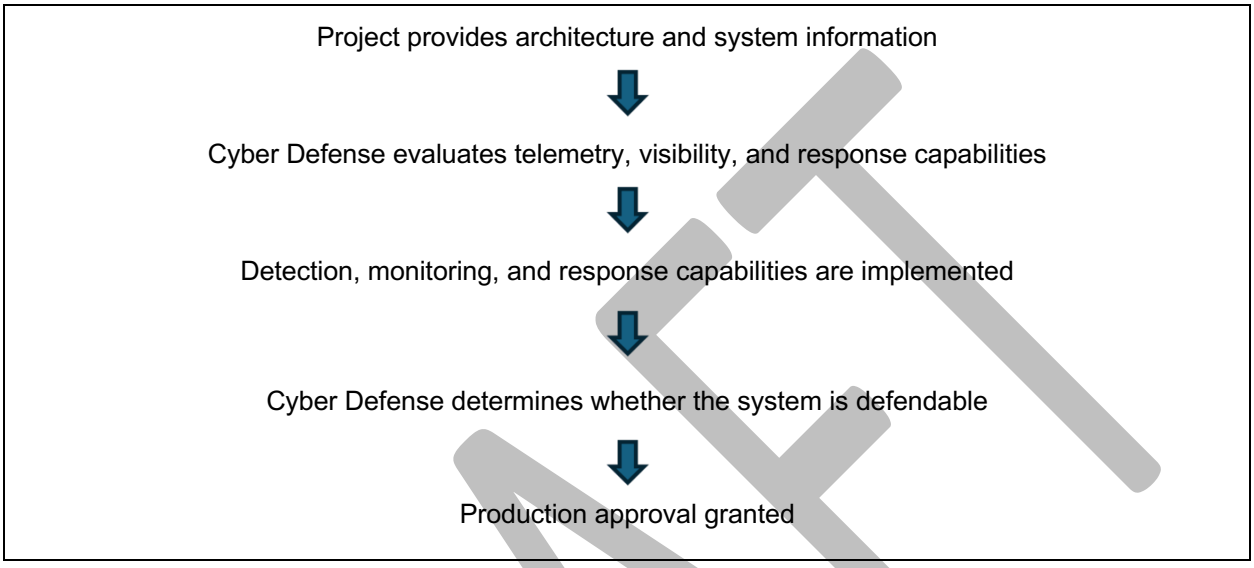

Work Package Inputs (Provided by the Project)

These inputs must be provided by the project team to enable Cyber Defense Enablement activities. These artifacts allow CD&I and Information Security teams to understand how the technology solution operates and how it will interact with the Cyber Defense ecosystem.

Input	Description	Owner
Solution Architecture	Architecture diagrams describing system components, trust boundaries, network flows, and integrations.	Project Architecture
Cyber Defense Architecture View (CDAV)	Architecture view identifying telemetry sources, defensive controls, monitoring points, and visibility gaps.	IS Architecture
Technology Stack Documentation	Infrastructure platforms, application frameworks, cloud services, identity providers, and security technologies used by the solution.	Engineering
Data Classification	Identification of sensitive data processed, stored, or transmitted by the solution.	Project / Security
External Integration Documentation	Third-party systems, APIs, SaaS platforms, and vendor integrations.	Project
Threat Model / Risk Assessment	Identified attack vectors and risk scenarios relevant to the solution.	IS Architecture / Security
Operational Ownership Model	Identification of operational support teams responsible for maintaining the solution.	SecOps / Business

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These inputs enable CD&I to determine how the solution can be monitored, detected, investigated, and defended.

Outputs to the Project

These outputs represent the formal outcomes of the Cyber Defense Enablement Work Package that are relevant to the project team.

Most Cyber Defense artifacts are internal operational assets and are not required by the project. The primary output provided to the project is the Cyber Defense readiness determination.

Output	Description	Owner
Cyber Defense Readiness Assessment	Determination that the technology solution has sufficient telemetry, detection capability, and response mechanisms to be defended by CD&I.	CD&I
Cyber Defense Production Approval	Formal approval indicating the system can enter production from a Cyber Defense monitoring and response perspective.	CD&I
Cyber Defense Gap Remediation Requirements (if applicable)	Required remediation actions the project must complete before Cyber Defense approval can be granted.	CD&I
Operational Monitoring Integration Confirmation	Confirmation that the solution is integrated with the Cyber Defense monitoring ecosystem.	CD&I SecOps

In most cases, Cyber Defense Enablement produces a single critical outcome for the project:

Approval (or conditional approval) for production deployment.

Internal Cyber Defense Deliverables

These artifacts are produced as part of the Cyber Defense Enablement Work Package but are internal to Cyber Defense and supporting security teams. They are necessary for operational defense but are not delivered to the project.

Internal Deliverable	Description	Owner
Cyber Defense Monitoring Requirements	Definition of telemetry, monitoring, and detection requirements needed to defend the solution.	CD&I
Telemetry Integration Requirements	Specifications for engineering teams to implement telemetry collection and forwarding.	CD&I / ISA
Detection Analytics	Detection rules, behavioral analytics, and correlation logic implemented within detection platforms.	CD&I / ISE / SecOps
Alerting and Automation Configuration	SOAR automations and alert routing workflows supporting incident response.	CD&I / ISE / SecOps

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Investigation Data Mapping	Identification of telemetry sources used for incident investigation and forensic analysis.	CD&I
Monitoring Runbooks	Operational procedures used by the Cyber Fusion Center for alert triage and monitoring.	CD&I
Incident Response Playbook Updates	Updates to incident response procedures incorporating the new technology solution.	CD&I
Defense Metrics and Dashboards	Monitoring metrics and reporting used to measure defensive effectiveness.	CD&I / IS Metrics

These artifacts enable Cyber Defense teams to **detect, investigate, and respond to threats affecting the solution once it enters production.**

Work Package Lifecycle

Phase 1 — Engagement (Project Input Phase)

Trigger Conditions
- New technology solution
- Major architecture change
- New external service integration

Project Inputs Required
- Cyber Defense Enablement Engagement Form
- Solution architecture documentation
- Cyber Defense Architecture View (CDAV)
- Technology stack documentation
- Data classification information
- External integration documentation

Activities
- CD&I engagement initiated
- Project inputs reviewed
- Cyber Defense Enablement scope defined
- Required internal workstreams identified

Output
- Cyber Defense Enablement engagement confirmed
- Initial defensive gap assessment

(No deliverables returned to the project at this stage)

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Phase 2 — Cyber Defense Architecture Review

Activities

- Review Cyber Defense Architecture View (CDAV)
- Identify telemetry sources
- Identify detection opportunities
- Identify investigation visibility
- Identify response and containment controls
- Identify visibility, detection, and response gaps

Internal Deliverables

- Cyber Defense monitoring requirements
- Telemetry integration requirements
- Detection engineering requirements

Output to Project

- Cyber Defense remediation requirements (if gaps must be addressed by the project)

Phase 3 — Telemetry & Engineering Integration

(Performed primarily by Engineering / SecOps)

Activities

- Telemetry pipelines implemented
- Event collection integrated
- Data normalization implemented
- Integration with enterprise monitoring platforms

Internal Deliverables

- Operational telemetry feeds
- Validated telemetry ingestion

Output to Project

- None (engineering work enabling cyber defense monitoring)

Phase 4 — Detection Engineering

(Performed by CD&I)

Activities

- Detection analytics developed
- Alerting logic implemented
- Automation workflows created
- Detection validation testing

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Internal Deliverables
- Detection analytics
- Alerting configuration
- SOAR automations

Output to Project
- None (These artifacts remain internal to Cyber Defense)

Phase 5 — Operational Enablement

(CD&I operational readiness)

Activities
- Monitoring runbooks created
- Incident response playbooks updated
- Operational dashboards configured
- Operational ownership confirmed

Internal Deliverables
- CFC monitoring runbooks
- IR playbook updates
- Operational metrics and dashboards

Output to Project
- None

Phase 6 — Cyber Defense Readiness Determination

Activities
- Telemetry validation
- Detection validation
- Operational readiness review
- Response capability validation

Output to Project
One of the following:
- Cyber Defense Production Approval
- Conditional Approval with Remediation Requirements
- Cyber Defense Non-Approval (Defensive Gaps Identified)

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Exit Criteria

The Cyber Defense Enablement work package is complete when:

- Required security telemetry is generated
- Telemetry is integrated with Cyber Defense monitoring platforms
- Detection analytics are implemented and validated
- Operational runbooks are established
- Incident response playbooks include the solution
- Operational ownership is confirmed

AND

- Cyber Defense production readiness determination has been issued

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Resource Model

Cyber Defense Enablement requires coordinated work across Cyber Defense, Information Security Architecture, Engineering, and Security Operations.

CD&I operates as an operational defense organization and does not maintain dedicated project engineering resources.

Projects introducing new technology solutions must therefore fund the resources required to implement Cyber Defense Enablement.

CD&I Responsibilities

Cyber Defense & Investigation is responsible for:

- defining monitoring and detection requirements
- identifying telemetry and visibility requirements
- developing detection analytics
- implementing alerting and automation
- creating monitoring runbooks
- updating incident response playbooks
- validating cyber defense readiness

CD&I does not implement telemetry pipelines or system integrations.

Information Security Architecture Responsibilities

Information Security Architecture is responsible for:

- producing the Cyber Defense Architecture View (CDAV)
- identifying defensive telemetry sources
- designing telemetry and monitoring integration architecture
- identifying defensive visibility gaps

Engineering Responsibilities

Information Security Engineering is responsible for:

- implementing telemetry collection pipelines
- integrating systems with enterprise monitoring platforms
- implementing required telemetry normalization and forwarding
- validating telemetry ingestion

Security Operations Responsibilities

Security Operations is responsible for:

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- operating production monitoring platforms
- maintaining telemetry pipelines
- supporting operational configuration of security platforms
- ensuring ongoing platform reliability

Project Responsibilities

Projects introducing new technology solutions are responsible for:

- providing required architecture documentation
- funding Cyber Defense Enablement work
- providing engineering resources required for telemetry integration
- addressing defensive gaps identified during architecture review

Roles and Responsibilities (RACI)

Activity	CD&I	IS Architecture	IS Engineering	SecOps	Project
Define monitoring requirements	A/R	C	C	C	I
Create Cyber Defense Architecture View	C	A/R	C	C	C
Telemetry integration design	C	A/R	C	C	C
Telemetry engineering implementation	C	C	A/R	C	R
Detection analytics development	A/R	C	C	C	I
Runbooks and playbooks	A/R	C	C	C	I
Operational support model	C	C	C	A/R	I
Production readiness validation	A/R	C	C	C	I

A = Accountable
R = Responsible
C = Consulted
I = Informed

Cost & Funding Model

Cyber Defense Enablement requires engineering, architecture, and operational work to ensure that technology solutions can be effectively monitored, detected, investigated, and defended by the Cyber Defense & Investigation (CD&I) organization. Projects introducing new technology solutions may have to fund the work required to enable Cyber Defense capabilities.

Funding Responsibility

Projects introducing or materially modifying technology solutions are responsible for funding the resources and technology required to implement Cyber Defense Enablement. These costs must be incorporated into project planning, estimation, and demand intake processes.

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Cost Categories

Projects may incur costs in several areas including:

- Architecture and design work required to produce the Cyber Defense Architecture View (CDAV)
- Engineering resources required to implement telemetry pipelines and monitoring integrations
- Detection engineering and analytics development
- Automation and orchestration configuration
- Monitoring platform ingestion capacity (e.g., SIEM or XDR)
- Telemetry storage and retention costs
- Security platform licensing associated with monitoring integrations

Estimated Effort

Typical Cyber Defense Enablement effort ranges depending on system complexity.

Work Component	Estimated Effort
Architecture review	1–2 weeks
Telemetry integration	3–6 weeks
Detection engineering	2–4 weeks
Runbooks and playbooks	1–2 weeks
Operational onboarding	1 week

Typical total effort:
6–12 weeks

Resource Funding

Projects may fund Cyber Defense Enablement work through:

- Internal engineering teams assigned to the project
- Contracted engineering or security specialists
- Consulting partners supporting the implementation effort

Operational Cost Considerations (TTB)

Technology solutions that increase the monitoring workload of the Cyber Defense organization may introduce ongoing operational costs. Projects must evaluate and account for potential increases in:

- Security telemetry volume
- Detection rule maintenance
- Telemetry pipeline maintenance
- Incident response workload

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Where necessary, additional operational funding or staffing must be considered as part of the total cost of ownership for the technology solution.

Cyber Defense Enablement Maturity Model

The Cyber Defense Enablement Maturity Model provides a clear framework for assessing how effectively a technology solution can be monitored, detected, investigated, and defended by the Cyber Defense & Investigation (CD&I) organization.

The model allows project teams, architects, and executive leadership to understand whether a system is operationally defensible against modern cyber threats.

The maturity model is not intended to measure compliance with security standards. Instead, it measures whether the system provides the operational capabilities required for real-world cyber defense.

Maturity Levels

Bronze (Minimal Defensibility)

Description
The technology solution provides limited or incomplete security visibility. Basic logs may exist and may be forwarded to central logging platforms, but the telemetry lacks the context required for effective detection or investigation.

Characteristics
- Basic logging exists but may lack security context
- Logs may be forwarded to SIEM or log repositories
- Limited or no detection analytics exist
- User activity may not be reliably attributable
- Investigation visibility is incomplete
- Incident response actions require manual intervention by system owners

Cyber Defense Impact
Attacks affecting the system may not be detected in a timely manner. Investigations may lack sufficient evidence to determine attacker behavior or impact.

Risk Profile
Systems at Bronze maturity create operational blind spots that attackers may exploit to establish persistence or expand access within the enterprise environment.

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Silver (Defendable)

Description
The technology solution integrates with the Cyber Defense ecosystem and provides sufficient telemetry and detection capability for CD&I to monitor and investigate suspicious activity.

Characteristics
- Security telemetry integrated with enterprise monitoring platforms
- Detection analytics implemented for common attack behaviors
- User activity can be attributed to identities or workloads
- Investigative telemetry available for incident response
- Monitoring runbooks and incident playbooks established
- Basic containment actions available (for example disabling accounts or revoking tokens)

Cyber Defense Impact
Suspicious activity can be detected and investigated. Cyber Defense teams can identify and respond to most attacks affecting the system.

Risk Profile
Systems at Silver maturity can be effectively defended by Cyber Defense operations and represent the minimum acceptable operational defense posture for production deployment.

Gold (Resilient Cyber Defense)

Description
The technology solution is fully integrated with enterprise Cyber Defense capabilities and supports advanced detection, investigation, and automated response.

Characteristics
- Comprehensive telemetry across identity, application, network, and data activity
- Advanced behavioral analytics and correlation detection
- Automated detection and response capabilities
- Real-time alerting integrated with enterprise detection platforms
- Full attack reconstruction capability for investigations
- Rapid containment controls integrated with security platforms

Cyber Defense Impact
Attacks are detected quickly and can be contained rapidly. Cyber Defense teams have the visibility and operational capability required to investigate complex multi-system attacks.

Risk Profile

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Systems at Gold maturity strengthen the overall enterprise cyber defense posture and significantly reduce the risk and impact of security incidents.

Enterprise Standard

All technology solutions deployed within the enterprise environment must achieve at least Silver maturity prior to production deployment.

Bronze-level systems create operational security blind spots and may expose the enterprise to unacceptable cyber risk.

Gold maturity represents the target state for systems that support critical enterprise operations or process sensitive data.

Production Approval Criteria

Cyber Defense Production Approval ensures that technology solutions entering the enterprise environment can be effectively defended by the Cyber Defense & Investigation (CD&I) organization.

Approval Criteria

A technology solution may be approved for production deployment by CD&I when the following conditions have been satisfied:

- Required security telemetry is generated by the system
- Telemetry is integrated with enterprise monitoring platforms
- Telemetry ingestion has been validated
- Detection analytics have been implemented and tested
- Cyber Defense monitoring runbooks are established
- Incident response playbooks include the technology solution
- Operational ownership of supporting platforms is confirmed
- Response and containment mechanisms are available where applicable

Validation Activities

Prior to approval, CD&I may perform validation activities including:

- Telemetry validation testing
- Detection validation exercises
- Incident investigation simulations
- Response capability verification

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Approval Outcomes

Following validation, CD&I will issue one of the following determinations:

- **Cyber Defense Production Approval:**
  The system is operationally defensible and may proceed to production.
- **Conditional Approval:**
  The system may proceed to production with defined remediation actions.
- **Non-Approval:**
  The system lacks sufficient defensive capability and must address identified gaps prior to production deployment.

Governance Requirement

Enterprise Governance Requirement

Cyber Defense Enablement is an enterprise governance requirement designed to ensure that technology solutions introduced into the enterprise environment can be actively defended against cyber threats.

Technology solutions that cannot be monitored, detected, investigated, and responded to by Cyber Defense create operational blind spots that attackers may exploit.

For this reason, all technology solutions introduced into the enterprise environment must undergo Cyber Defense Enablement review.

Production Deployment Requirement

A technology solution may not enter production until Cyber Defense Enablement activities have been completed and CD&I has issued a production readiness determination.

This requirement ensures that:

- Cyber Defense maintains visibility into system activity
- Threats affecting the system can be detected
- Incidents can be investigated and contained
- The enterprise maintains a defensible security posture

Key Principle

Cyber Defense Enablement does not exist to produce artifacts for projects. It exists to ensure that the Cyber Defense organization can effectively defend the technology solution once it is deployed. Cyber Defense Enablement ensures that enterprise technology solutions are operationally defensible.

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The primary output of the Cyber Defense Enablement Work Package is: A determination that the system can be monitored, detected, investigated, and defended by Cyber Defense. Without this capability, the system should not proceed to production deployment.

Exception Handling

If a technology solution cannot achieve Silver maturity prior to deployment, a formal risk exception must be submitted and approved through the enterprise risk governance process.

The exception must:

- document the cyber defense capability gaps
- describe the operational risks created by the gaps
- define compensating controls where applicable
- include a remediation plan and timeline to achieve Silver maturity

Risk exceptions must be approved by the CTISO or designated risk authority prior to production deployment.

Note: Approval of a risk exception does not eliminate the cyber defense capability gap. It formally transfers ownership of the operational risk to executive leadership.

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Appendix A – Engagement Form

Purpose

This form initiates engagement with the Cyber Defense & Investigation (CD&I) team to determine whether the Cyber Defense Enablement Work Package (CDEP) is required for a technology solution.

Completion of this form provides CD&I with the information necessary to evaluate how the solution will be monitored, detected, investigated, and defended once deployed.

Project Information

Project Name:

Project ID / Intake ID:

Project Manager:

Business Owner:

Solution Architect:

Engineering Lead:

Target Production Date:

Project Type: ☐ New System ☐ Major Enhancement ☐ Infrastructure Change ☐ Vendor Solution ☐ Other

Technology Solution Overview

Provide a brief description of the technology solution being delivered:

_____

_____

_____

_____

Architecture and Exposure

Does the solution introduce a new application or platform?	☐ Yes ☐ No
Will the solution be internet accessible?	☐ Yes ☐ No
Will the solution process or store sensitive data?	☐ Yes ☐ No
Does the solution integrate with external vendors or APIs?	☐ Yes ☐ No
Will the solution be deployed in cloud infrastructure?	☐ Yes ☐ No
Does the solution introduce new authentication mechanisms?	☐ Yes ☐ No

Cyber Defense Visibility

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Does the solution generate security-relevant telemetry or logs? ☐ Yes ☐ No ☐ Unknown

Are logs or telemetry planned to be forwarded to monitoring platforms? ☐ Yes ☐ No ☐ Unknown

Does the system support user activity attribution? ☐ Yes ☐ No ☐ Unknown

Can the system support investigation of user or data activity? ☐ Yes ☐ No ☐ Unknown

Response and Containment Capability

Can compromised accounts or credentials be disabled or revoked? ☐ Yes ☐ No ☐ Unknown

Can system access be restricted during an incident? ☐ Yes ☐ No ☐ Unknown

Are administrative controls available to stop malicious activity? ☐ Yes ☐ No ☐ Unknown

Supporting Documentation

Please attach or provide links to the following documentation if available:

- Solution Architecture Diagram
- Network / Data Flow Diagram
- Threat Model or Risk Assessment
- Vendor Architecture Documentation

Submission

Submit the completed form to Cyber Defense & Investigation (CD&I).

Upon submission:

1. CD&I and Information Security Architecture will review the request.
2. Cyber Defense Enablement scope will be determined.
3. The Cyber Defense Enablement Work Package will be initiated if required.

Important Note

Cyber Defense Enablement ensures that technology solutions can be actively defended against cyber threats.

A system that cannot be monitored, detected, investigated, and defended by Cyber Defense is not operationally defensible and may not be approved for production deployment.

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